

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

Max. Marks: 100

APRIL - MAY 2016
End Semester Exam

Duration: 3hrs

Class: SY B.Tech

Semester: III

Name of the Course: Theory of Computers

Course Code: UITC305

Instructions:

- (1) All Questions are Compulsory
- (2) Answer to each new question to be started on a fresh page.
- (3) Figures in brackets on the right hand side indicate full marks

Question No.		Max. Marks
Q1(a)	i. List and explain the limitations of finite Automata. ii. Write a short note on Ambiguity resolution. iii. Draw NFA with ϵ moves $((10+011)^*+11)^*+0101$	[4m] [4m] [2m]
Q1(b)	i. Find CFG for the regular expression $(110+11)^*(10)^*$. ii. Build CFG for the language $L=\{0^i 1^j 2^k \mid j \geq i+k\}$ Or Design a mealy machine for the language $(0+1)^*(00+11)$ and convert the Mealy machine to Moore machine.	[5m] [5m] [10m]
Q2(a)	i) Explain Church Turing Thesis ii) Construct the finite automata Equivalent to RE. $(0+1)^*(00+11)(0+1)^*$	[5m] [5m]
2(b)	Consider the grammar :- $S \rightarrow 0B \mid 1A$ $A \rightarrow 0 \mid 0S \mid 1AA$ $B \rightarrow 1 \mid 1S \mid 0BB$ For the string 00110101 find the following :- (i) Leftmost derivation (ii) Rightmost derivation (iii) Parse tree.	[10m]

3(a)	(i) Design FA which accepts even number of 0's and even number of 1's .	[5m]															
	(ii) Design FA to accept L where all string in L are such that total number of a's in them are divisible by 3 .	[5m]															
	Or Design Turing machine to recognize $L = \{a^n b^n c^n \mid n \geq 1\}$	[10m]															
3(b)	(i) Explain the closure properties of Context free Languages.	[5m]															
	(ii) Differentiate between Deterministic and non deterministic PDAs	[5m]															
4(a)	(i) Explain Variation in Turing machine	[5m]															
	(ii) Explain Chomsky hierarchy.	[5m]															
	Or Construct PDA for given CFG $S \rightarrow aSb$ $S \rightarrow ab$	[10m]															
4(b)	Express the following language in CNF and GNF $L = \{a^n b^n \mid n \geq 1\}$	[10m]															
5(a)	(i) Using pumping Lemma show that the following language is not regular $L = \{0^n 1^{n+1} \mid n > 0\}$	[5m]															
	(ii) Explain halting problem in detail.	[5m]															
5(b)	Show that the following language is not context free. $L = \{a^n b^m c^p \mid n \geq 0, m = 2n \text{ and } p = m+n\}$ Or Convert the given NFA to DFA . <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>State\input</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <td>q0</td><td>{q0, q1}</td><td>q0</td></tr> <tr> <td>q1</td><td>q2</td><td>q1</td></tr> <tr> <td>q2</td><td>q3</td><td>q3</td></tr> <tr> <td>q3*</td><td>Ø</td><td>q2</td></tr> </tbody> </table> *final state	State\input	0	1	q0	{q0, q1}	q0	q1	q2	q1	q2	q3	q3	q3*	Ø	q2	[10m]
State\input	0	1															
q0	{q0, q1}	q0															
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